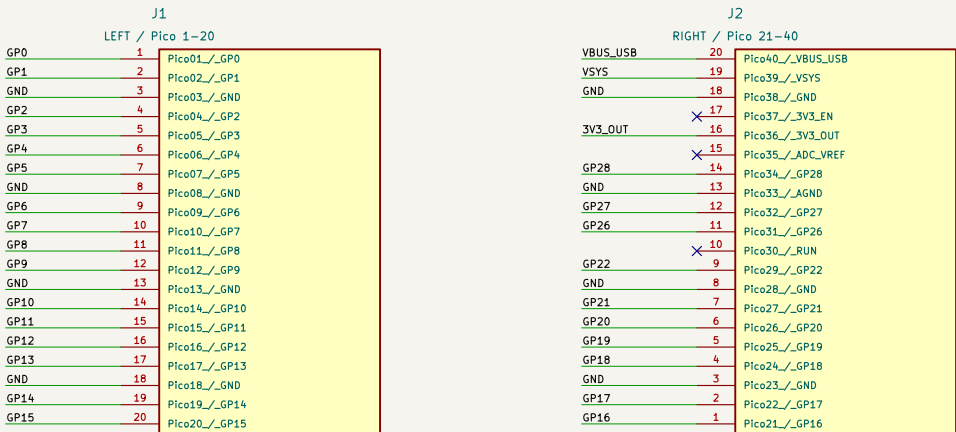


MAKE MUSIC – PROVISIONAL PICO CARRIER

Editable KiCad 10 schematic | Module order, parts, power and mechanical fit require primary integration

REMOVABLE PICO: TOP VIEW, USB AT TOP

Two 1x20 female sockets: 2.54 mm pitch, 17.78 mm row spacing. Carrier contacts are passive.



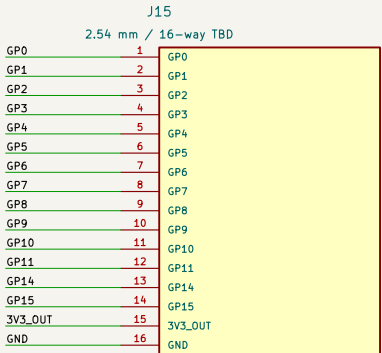
J1 pin 1 = Pico 1 (USB end). J2 pin 1 = Pico 21 (opposite USB).

Right socket shown pin 20 at USB end = Pico 40; match footprint pad numbers before layout.

NC: Pico 30 RUN, 35 ADC_VREF, 37 3V3_EN. On-module defaults retained.

Pico 33 AGND tied to GND for digital touch use. GP23–25/29 are internal, not expansion.

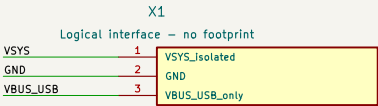
FREE GPIO EXPANSION – ORDER PROVISIONAL



GP0/GP1: optional I2C SDA/SCL.

No I2C pull-ups fitted until bus/device choice.

POWER REVIEW HANDOFF



X1 is an integration contract, not a connector.

VSYS: protected/switched/source-isolated supply.

VBUS_USB: Pico USB rail for power review only.

Do not join VBUS_USB to VSYS on carrier.

3V3_OUT comes from Pico pin 36 regulator.

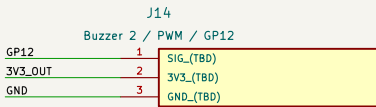
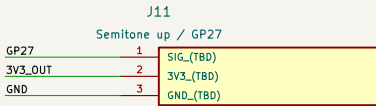
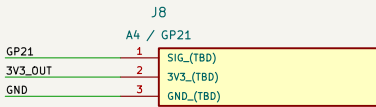
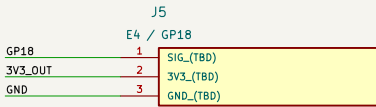
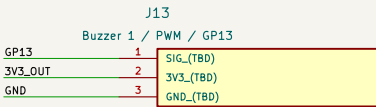
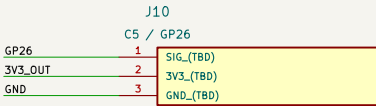
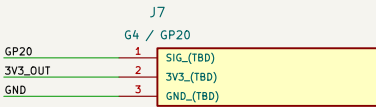
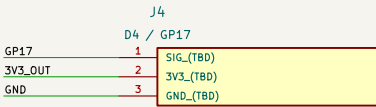
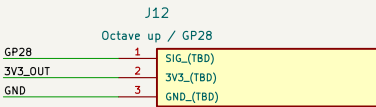
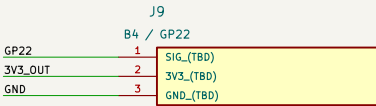
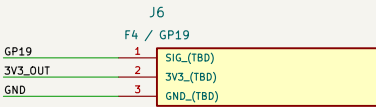
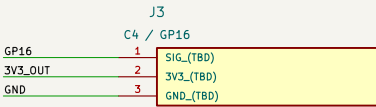
No battery, charger or protection circuit fitted.

Raw battery must NEVER feed GPIO or 3V3_OUT.

TEN TOUCH / TWO BUZZER MODULE INTERFACES

ALL J3–J14: assumed 1=signal, 2=3V3, 3=GND. Verify actual module pin order before assembly.

Footprints unassigned. Only 3.3 V-compatible modules with suitable output/driver circuits may be used.



SIGNAL / POWER LIMITS TO RESOLVE

Touch candidate: HiLetgo/TTP223 at 3.3 V. Firmware needs direct active-high (TOG=0, AHLB=0).

Buzzers: signal is a PWM logic input to a compatible driver module. Do not drive bare coils directly.

GP12 + GP13 share a PWM slice/frequency. Baseline firmware plays both at the same pitch.

Budget Pico 3V3_OUT current including 10 touch modules, 2 buzzer drivers and expansion loads.

Socket MPN/footprints, USB/BOOTSEL access, insertion height, module and battery clearances TBD.

Reference image may be active-low (AHLB=1). Confirm configuration and pin order; see review.

REVIEW ONLY – NOT FOR FABRICATION

ERC validates carrier connectivity only; off-board electronics and unresolved power design are not modeled.

Sources: Raspberry Pi Pico/Pico 2 datasheets, Figure 2 pin allocation; see ../reviews/schematic.md.

Review only – parts and power unresolved

Sheet: /
File: make_music.kicad_sch

Title: Make Music – provisional Pico carrier

Size: A3	Date: 2026-10-07	Rev: P0
KiCad E.D.A. 10.0.6		Id: 1/1